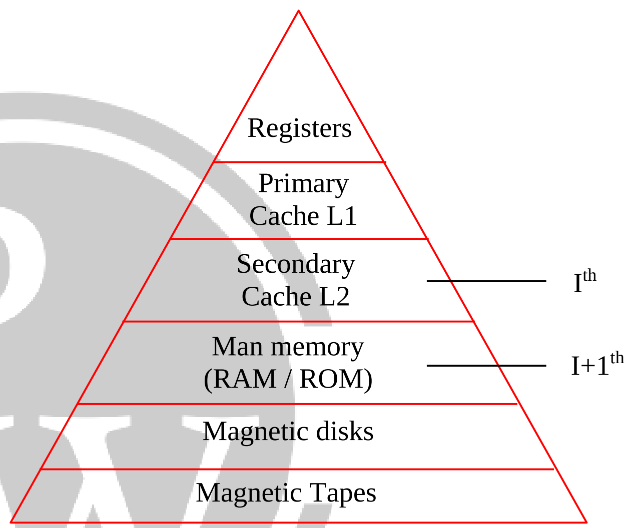
5 Memory Interfacing

5.1 Memory Hierarchy

- The memory hierarchy system consists of all storage devices.

  The purpose of memory hierarchy is to bridge the speed mismatch between the fastest processor to the slowest memory component at reasonable cost.



- In the structure of memory hierarchy I^{th} level memory is physically positioned higher than $(I + 1)^{H}$ level memory.
- Let Ti, Si, Ci, & Fi are the access time, size, cost per bit and frequency of references to the I^{th} level memory. Therefore

$$T_i < T_i + 1$$

$$S_i < S_i + 1$$

$$C_i > C_i + 1$$

$$f_i > f_i + 1$$

- Since same data presents at different levels. I^{th} level memory is the subset of $I\ i + 1^{th}$ level.

  i.e. $\quad I\ i \subset I\ i + 1$

5.2 Memory Characteristics

Location: Memory can be placed in 3 locations like registers, main memory, Auxiliary (or) secondary storage.

Capacity: Word size, number of words i.e. capacity = number of words * word size.

Unit of transfer: Maximum number of bits that can be read or written (blocks, bytes…)

Access method: Random or sequential

Performance: Access time, memory cycle time, transfer rate, physical type.

Physical: Volatile / non volatile

erasable / non erasable

	Serial Access		Random Access
(1)	Memory is organized into units of data called records/blocks accessed sequentially	**(1)**	Each storage location has an address uniquely
(2)	Access time depends on position of storage location	**(2)**	Access time is independent of storage location
(3)	Slower to Access	**(3)**	Faster to access
(4)	Cheaper	**(4)**	Costly relatively
(5)	Nonvolatile memories	**(5)**	May be relative or non
(6)	Example: Magnetic tapes	**(6)**	Example: Magnetic disks.

- When processor reads I^{th} level memory, if it is found in that level, his will occur otherwise it will be fault.

5.2.1. In a Two – Level Memory System

Case – I: When fault occurs, in L1 reads from L2 (Proxy Hierarchy)

T_1, S_1, H_1, C_1

L_1

T_2, S_2, C_2

L_2

$$T_{avg} = H_1 * T_1 + (1–H_1) * T_2$$

Case – II: When fault occurs in L_1, must be brought from L2 to L1 memory. (**strict hierarchy**)

$$T_{avg} = H_1 * T_1 + (1–H_1) * (T_2 + T_1)$$

5.2.2. In a Three – Level Memory System

Case – I: When fault occurs in one level, then reads from its down level.

T_1, S_1, H_1, C_1

L_1

T_2, S_2, H_2, C_2

L_2

T_2, S_2, H_2, C_2

L_3

$$T_{avg} = H_1 * T_1 + (1 - H_1) * H_2 * T_2 + (1 - H_1) * (1 - H_2) * T_3$$

Case – II: When fault occurs, must access from L1.

$$T_{avg} = H_1 * T_1 + (1 - H_1)(H_2)(T_2 + T_1) + (1 - H_1)(1 - H_2)(T_3 + T_2 + T_1)$$

* Average cost per bit

$$C_{Avg} = \frac{C_1S_1 + C_2S_2}{S_1 + S_2} \qquad \text{(Two - level)}$$

$$C_{Avg} = \frac{C_1S_1 + C_2S_2 + C_3S_3}{S_1 + S_2 + S_3} \qquad \text{(Three - level)}$$